

Assessment Schedule – PRACTICE ASSESSMENT B

Calculus: Apply differentiation methods in solving problems (91578)

Evidence Statement

Q	Expected Coverage	Achievement (u)	Achievement with Merit (r)	Achievement with Excellence (t)
ONE (a)	$f'(x) = 4(-3x^2 + 2)(6 - x^3 + 2x)^3.$	• Correct derivative.		
(b)	$\frac{dx}{d\theta} = 3 \sec^2(\theta).$ $\frac{dy}{d\theta} = 4 \sin(\theta) \cos(\theta).$ $\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$ $= (4 \sin(\theta) \cos(\theta)) \cdot \left(\frac{\cos^2(\theta)}{3}\right) =$ $\frac{4 \sin(\theta) \cos^3(\theta)}{3}.$ When $\theta = \frac{\pi}{4}, \frac{dy}{dx} = \frac{4 \sin(\frac{\pi}{4}) \cos^3(\frac{\pi}{4})}{3} = \frac{1}{3}.$	• Correct rate of change found, with evidence of derivative.		
(c)	$f'(x) = \frac{xe^x - e^x}{x^2} + 2.$ When $x = 1, f'(x) = e - e + 2 = 2.$ When $x = 1, f(x) = e + 2.$ $y - y_1 = m(x - x_1),$ $y - (e + 2) = 2(x - 1),$ $y = 2x - 2 + e + 2 = 2x + e.$ Hence $a = 2$ and $b = 0.$	• Correct expression for $f'(x)$ AND $f'(x)$ evaluated when $x = 1.$	• Correct value of a and b found, with evidence of derivative.	
(d)	$f(2) = \frac{2^2}{2-1} = 4, f(5) = \frac{5^2}{5-1} = \frac{25}{4},$ $\frac{f(b)-f(a)}{b-a} = \frac{\frac{25}{4}-4}{5-2} = \frac{3}{4}.$ $f'(x) = \frac{2x(x-1)-x^2}{(x-1)^2} = \frac{x^2-2x}{(x-1)^2}.$ Solving $\frac{x^2-2x}{(x-1)^2} = \frac{3}{4},$ $(x-3)(x+1) = 0,$ $x = 3$ or $x = -1,$ but $x \neq 1$ since $2 \leq x \leq 5.$ $f(3) = \frac{9}{2}.$ Hence the point that is guaranteed to exist is $(3, \frac{9}{2}).$	• Correct expression for $f'(x).$	• Correct coordinates of the point found, with evidence of derivative.	

(e)	$f'(x) = -e^{-x}(x^2 + kx + 1) + e^{-x}(2x + k),$ To find stationary points, $f'(x) = 0,$ $-x^2 - (k - 2)x + (k - 1) = 0.$ The curve has exactly one stationary point, $\Delta = b^2 - 4ac = 0,$ $(k - 2)^2 + 4(k - 1) = 0,$ $k^2 = 0,$ hence $k = 0.$ $f'(x) = -e^{-x}(x^2 + 1) + 2xe^{-x}.$ $f''(x) = e^{-x}(x^2 + 1) - e^{-x}(2x) + 2e^{-x} - 2xe^{-x}.$ To find points of inflection, $f''(x) = 0,$ $x^2 - 4x + 3 = 0,$ $\Delta = b^2 - 4ac = 4^2 - 4 \times 3,$ $\Delta = 4 > 0,$ $\therefore$ the curve always has points of inflection.	• Correct expression for $f'(x).$	• Correct expression for both $f'(x)$ and $f''(x).$	E7 (T1) $k = 0$ found AND $x^2 - 4x + 3 = 0$ found, with evidence of both $f'(x)$ and $f''(x)$ OR Correct solution, but with one minor error. E8 (T2) Complete proof, with evidence of a calculus method.
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NØ	N1	N2	A3	A4	M5	M6	E7	E8
No response; no relevant evidence.	ONE partial solution.	1u	2u	3u	1r	2r	T1	T2

Q	Expected Coverage	Achievement (u)	Achievement with Merit (r)	Achievement with Excellence (t)
TWO (a)	$f'(x) = 1 - \frac{\operatorname{cosec}(x) - x(-\operatorname{cosec}(x)\cot(x))}{\operatorname{cosec}^2(x)}$, OR $f'(x) = 1 - (\sin(x) + x \cdot \cos(x))$.	Correct derivative.		
(b)	$C'(t) = 24\left(\frac{3}{3t+1}\right)(\ln(3t+1))$. $C'(5) = \frac{72 \ln 16}{16} = 4.5 \ln 16$, ≈ 12.48 (gram per day).	Correct rate of change, with evidence of derivative.		
(c)	$f'(x) = \cos^2(\pi x) + x(-2\pi \sin(\pi x) \cos(\pi x))$, $= \cos^2(\pi x) - 2\pi x \sin(\pi x) \cos(\pi x)$, $f'(1) = 1$. $g'(x) = 2kx - 3$, $g'(1) = 2k - 3$. The two normals are perpendicular so $\left(-\frac{1}{f'(x)}\right)\left(-\frac{1}{g'(x)}\right) = -1$, which can be simplified to $f'(x)g'(x) = -1$. Solve $1(2k - 3) = -1$, $k = 1$. $g(x) = x^2 - 3x$, $g(1) = -2$, $g'(1) = 2x - 3 = -1$. Equation of the tangent: $y - y_1 = m(x - x_1)$, $y - (-2) = -1(x - 1)$, $y = -x - 1$.	Correct expression for both $f'(x)$ and $g'(x)$.	Correct equation of tangent found, with evidence of derivatives.	
(d)	Volume of a cone $V = \frac{1}{3}\pi r^2 h$. $\frac{r}{h} = \frac{10}{40} = \frac{1}{4}$, $r = \frac{h}{4}$. Volume of water filled $V = \frac{1}{3}\pi\left(\frac{h}{4}\right)^2 h = \frac{\pi h^3}{48}$. $\frac{dV}{dh} = \frac{\pi h^2}{16}$. Given $\frac{dV}{dt} = 18$, The rate at which the depth of water is increasing $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt} = \frac{16 \times 18}{\pi h^2}$. When $h = 15$, $\frac{dh}{dt} = \frac{288}{225\pi} \approx 0.407 \text{ cm s}^{-1}$.	Correct expression for $\frac{dV}{dh}$.	Correct rate of change, with evidence of derivative.	
(e)	Let the height of the ball above the ground be h . $\tan \theta = \frac{h-1}{x}$, $h(\theta) = 1 + x \tan \theta$, $h(\theta) = 1 + (3 \sin \theta \cos^2 \theta)\left(\frac{\sin \theta}{\cos \theta}\right)$, $h(\theta) = 1 + 3 \sin^2 \theta \cos \theta$. $h'(\theta) = 6 \sin \theta \cos^2 \theta - 3 \sin^3 \theta$. To find maximum, $h'(\theta) = 0$, $3 \sin \theta (2 \cos^2 \theta - \sin^2 \theta) = 0$, If $\sin \theta = 0$, $\theta = 0$, this will not be the maximum. $\therefore 2 \cos^2 \theta - \sin^2 \theta = 0$, $\tan^2 \theta = 2$, $\theta = 0.9553$. $h(0.9553) = 2.15 \text{ metres} < 2.2$. $\therefore$ The ball will never pass completely over Ben.	Finds expression for $h'(\theta)$, but with one minor error.	Correct expression for $h'(\theta)$ AND Sets $h'(\theta) = 0$.	E7 (T1) Correct solution, but with one minor error. E8 (T2) Correct maximum height and correct conclusion, with evidence of a calculus method.

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No response; no relevant evidence.	ONE partial solution.	1u	2u	3u	1r	2r	T1	T2

Q	Expected Coverage	Achievement (u)	Achievement with Merit (r)	Achievement with Excellence (t)
THREE (a)	(i) $x < -2, x = 3$ (ii) $x = 1, x = 4$ (iii) Does not exist	• TWO out of three responses correct.		
(b)	$f'(x) = \frac{4e^{\frac{1}{8}x^2} - 4x(\frac{1}{4}x)e^{\frac{1}{8}x^2}}{\left(e^{\frac{1}{8}x^2}\right)^2}.$ When $f'(x) = 0,$ $e^{\frac{1}{8}x^2}(4 - x^2) = 0.$ $e^{\frac{1}{8}x^2} \neq 0 \therefore 4 - x^2 = 0.$ $x = \pm 2.$	• Correct x -coordinates found, with evidence of derivative.		
(c)	Given that the value of y always increases as the same rate as the value of $x,$ $\frac{dy}{dx} = 1.$ $\frac{dy}{dx} = \frac{(kx+1)(2x+2) - k(x^2+2x+1)}{(kx+1)^2} = 1.$ $kx^2 + 2x + (2 - k) = k^2x^2 + 2kx + 1,$ hence $k = 1.$	• Correct expression for $\frac{dy}{dx}.$	• Correct value of k found, with evidence of derivative.	
(d)	$\frac{dx}{dt} = 3 \times (4t^3) \times (t^4 - 1)^2,$ $\frac{dy}{dt} = (2t - 6)e^t + (t^2 - 6t + 10)e^t.$ $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = \frac{e^t(2t-6+t^2-6t+10)}{12t^3(t^4-1)^2}.$ The tangent is parallel to x -axis so $\frac{dy}{dx} = 0.$ Hence $e^t(2t - 6 + t^2 - 6t + 10) = 0,$ $e^t(t - 2) = 0,$ $t = 2.$ When $t = 2, y = (4 - 12 + 10)e^2,$ $y = 2e^2.$ $y - y_1 = m(x - x_1),$ $y - 2e^2 = 0(x - x_1),$ Therefore the equation of the tangent is: $y = 2e^2.$	• Correct expression for $\frac{dy}{dx}.$	• Correct equation of the tangent found, with evidence of derivative.	

(e)	$2x^2 - kx + 3 > \frac{k}{x},$ $2x^3 - kx^2 + 3x > k,$ $2x^3 + 3x > k(x^2 + 1),$ $k < \frac{2x^3+3x}{x^2+1}.$ Hence k must be smaller than the minimum value of $\frac{2x^3+3x}{x^2+1}.$ Let $h(x) = \frac{2x^3+3x}{x^2+1},$ $h'(x) = \frac{(x^2+1)(6x^2+3)-(2x^3+3x)(2x)}{(x^2+1)^2},$ $h'(x) = \frac{2x^4+3x^2+3}{(x^2+1)^2}.$ Clearly $h'(x)$ must $> 0,$ since $1 \leq x \leq 2, x = 1$ will minimise $h(x).$ When $x = 1, \frac{2x^3+3x}{x^2+1} = \frac{5}{2},$ $\therefore k < \frac{5}{2}.$	• Correct derivative ($h'(x)$) Does not need to be simplified.	• Correct derivative ($h'(x)$) AND Recognises $h'(x) > 0$ OR $x = 1,$ though justification may be incomplete.	E7 (T1) Correct solution, but with one minor error. E8 (T2) Complete proof with full justification and evidence of a calculus method.
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No response; no relevant evidence.	ONE partial solution.	1u	2u	3u	1r	2r	T1	T2

Cut Scores (2025)

Not Achieved	Achievement	Achievement with Merit	Achievement with Excellence
0 – 7	8 – 13	14 – 19	20 – 24

Cut Scores (2024)

Not Achieved	Achievement	Achievement with Merit	Achievement with Excellence
0 – 7	8 – 12	13 – 18	19 – 24